SIGNHIFY STUDIO // PRODUCTION ENGINEERING BLUEPRINT

Zero-PC Cloud Automation Architecture

How to Run Autogram 24/7 in the Cloud for $0 Using Free Developer Infrastructure

ENGINE:	STUDIO:	HARDWARE:	CADENCE:
Autogram v5.2	signhify.dpdns.org	$0 Required	4 Drops Daily

1. Executive Summary & Strategy

This blueprint details how to transition Autogram from running locally on your workstation to an enterprise-grade, 100% autonomous cloud pipeline that operates 24/7 with zero PC uptime. By chaining five free cloud developer tiers, your content publishing machine generates high-retention carousels, video reels, and interactive stories while you sleep.

2. The Zero-Dollar Cloud Tech Stack

Service	Free Allowance	Role in Autogram	Cost
GitHub Actions	2,000 Linux min/mo (Private) / Unlimited (Public)	Executes pipeline, runs Playwright & Remotion, commits memory	**$0.00**
Cloudflare Workers	100,000 req/day, 10ms CPU	Exact-second cron triggers to eliminate GitHub Actions delay	**$0.00**
Cloudflare R2	10 GB storage/mo, $0 egress fees	Zero-cost media hosting for Meta Graph API container ingestion	**$0.00**
Render.com	750 free instance hours/mo	Hosts dashboard API and emergency failover webhook endpoint	**$0.00**
ManyChat Cloud	Free tier (up to 1,000 contacts)	Cloud-native comment listener & follow-gated DM automations	**$0.00**

> 💡 **Why Cloudflare R2 Beats AWS S3**
>
> Meta's Graph API must download every 1080×1350 image slide and 60fps MP4 reel directly from a public URL. AWS S3 charges $0.09/GB for data egress. Cloudflare R2 has **zero egress bandwidth fees**, saving hundreds of dollars per year at high publishing volumes.

3. GitHub Actions Master Workflow: `daily-autopilot.yml`

Save this configuration file in your repository at `.github/workflows/daily-autopilot.yml`:

```yaml
name: Autogram 24/7 Cloud Autopilot

on:
  schedule:
    # 4 Daily Publishing Drops (Asia/Kolkata: UTC + 5:30)
    - cron: '0 4 * * *'   # 09:30 AM IST (04:00 UTC) — Morning High-Signal Story
    - cron: '0 6 * * *'   # 11:30 AM IST (06:00 UTC) — Midday Prime Carousel
    - cron: '0 12 * * *'  # 05:30 PM IST (12:00 UTC) — Evening Tech Radar Story
    - cron: '0 14 * * *'  # 07:30 PM IST (14:00 UTC) — Evening Prime Reel / Carousel
  repository_dispatch:
    types: [publish-slot, trigger-autopilot]
  workflow_dispatch:

jobs:
  autopilot:
    runs-on: ubuntu-latest
    timeout-minutes: 25
    permissions:
      contents: write

    steps:
      - name: 1. Checkout Repository
        uses: actions/checkout@v4
        with:
          fetch-depth: 0

      - name: 2. Set Up Python 3.12
        uses: actions/setup-python@v5
        with:
          python-version: "3.12"
          cache: "pip"

      - name: 3. Install Core Python Dependencies
        run: pip install -r requirements.txt

      - name: 4. Install Playwright Headless Chromium & System Fonts
        run: |
          playwright install --with-deps chromium
          sudo apt-get update && sudo apt-get install -y fonts-inter ffmpeg

      - name: 5. Execute Autonomous Publishing Engine
        env:
          AUTOGRAM_OWNER_KEY: ${{ secrets.AUTOGRAM_OWNER_KEY || 'autogram_owner_vip_2026' }}
          GEMINI_API_KEY: ${{ secrets.GEMINI_API_KEY }}
          OPENROUTER_API_KEY: ${{ secrets.OPENROUTER_API_KEY }}
          CLOUDFLARE_API_TOKEN: ${{ secrets.CLOUDFLARE_API_TOKEN }}
          CLOUDFLARE_ACCOUNT_ID: ${{ secrets.CLOUDFLARE_ACCOUNT_ID }}
          IG_USER_ID: ${{ secrets.IG_USER_ID }}
          IG_ACCESS_TOKEN: ${{ secrets.IG_ACCESS_TOKEN }}
          PUBLIC_CDN_BASE: ${{ secrets.PUBLIC_CDN_BASE }}
          DRY_RUN: 'false'
```

```
        run: python orchestrator.py --run-all

      - name: 6. Persist Content Memory & Anti-Repetition Ledger
        if: always()
        run: |
          git config --global user.name "autogram-cloud-bot"
          git config --global user.email "bot@signhify.dpdns.org"
          git pull --rebase origin main || true
          git add data/content-memory.json data/schedule.json autopilot.db || true
          git diff --cached --quiet || git commit -m "chore(cloud): sync memory [skip ci]"
          git push origin main || true
```

4. Cloudflare Worker: Zero-Jitter Cron Dispatcher

GitHub's native scheduled crons can occasionally queue for 5 to 20 minutes during peak hours. Deploy this lightweight Cloudflare Worker with a Cron Trigger to fire an instant `repository_dispatch` event to GitHub at the exact second:

```
// Cloudflare Worker: autogram-cron-dispatcher.js
export default {
  async scheduled(event, env, ctx) {
    const GITHUB_REPO = "YourUsername/Autogram";
    const GITHUB_TOKEN = env.GH_PAT_TOKEN; // GitHub Personal Access Token

    const response = await fetch(`https://api.github.com/repos/${GITHUB_REPO}/dispatches`, {
      method: "POST",
      headers: {
        "Accept": "application/vnd.github.v3+json",
        "Authorization": `Bearer ${GITHUB_TOKEN}`,
        "User-Agent": "Autogram-Cloudflare-Cron"
      },
      body: JSON.stringify({
        event_type: "publish-slot",
        client_payload: { slot: event.cron, timestamp: new Date().toISOString() }
      })
    });
    console.log(`Dispatched GitHub Actions Autopilot: HTTP ${response.status}`);
  }
};
```

5. Required GitHub Secrets Checklist

Secret Key	Description	Value / Source
`AUTOGRAM_OWNER_KEY`	HMAC-SHA256 license bypass	`autogram_owner_vip_2026`
`GEMINI_API_KEY`	Google Gemini AI key	Google AI Studio
`OPENROUTER_API_KEY`	Multi-LLM fallback chain	OpenRouter.ai
`IG_USER_ID`	Instagram Business Account ID	Meta Developer Console
`IG_ACCESS_TOKEN`	60-Day Long-Lived Token	Meta Graph API Explorer
`CLOUDFLARE_API_TOKEN`	Cloudflare R2 Bucket Token	Cloudflare Dashboard → R2
`PUBLIC_CDN_BASE`	Public Media Base URL	`https://cdn.signhify.dpdns.org`

6. Final Deployment Checklist

1. **Push Local Code to GitHub:** Ensure all files in `d:\Autogram` are pushed to your remote `main` branch.
2. **Populate GitHub Secrets:** Add the 7 keys from the table above into GitHub Repository Settings.
3. **Enable Workflow Write Permissions:** Under *Settings → Actions → General → Workflow permissions*, select **"Read and write permissions"** so the cloud runner can commit the updated anti-repetition memory back to Git.
4. **Trigger Initial Test Run:** Go to *Actions → Autogram 24/7 Cloud Autopilot → Run workflow*.
5. **Power Down Your Workstation:** Your machine can now be turned off. The cloud infrastructure will handle 100% of generation, rendering, uploading, and publishing autonomously.